

Class 4 Math

Premium Single-Column Workbook

Featuring 100+ Advanced Master Problems & Plenty of Space to Think

Chapter 1: Large Numbers

SECTION A: CORE CONCEPTS

1. Write the number name for 4,567,890 in the International System.

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2. What is the place value and face value of 8 in 9,875,321?

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3. Write $400,000 + 50,000 + 6,000 + 200 + 90 + 1$ in standard form.

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4. Round 54,672 to the nearest thousand.

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5. Arrange in ascending order: 45,678; 45,876; 45,687; 45,786.

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SECTION B: REAL WORLD APPLICATIONS

1. The population of San Jose is approximately 971,233. Round this number to the nearest ten thousand.

2. A factory produced 125,450 packets of Crunchley makhana in January and 132,600 packets in February. Which month had higher production, and write that number in expanded form.

3. Mukesh's new software code has 1,204,500 lines. What is the digit in the hundred-thousands place?

Chapter 2: Advanced Addition & Subtraction

SECTION A: CORE CONCEPTS

1. Add: $45,678 + 32,194$

2. Add: $105,402 + 89,500 + 12,045$

3. Subtract: $90,000 - 45,672$

4. Subtract 12,450 from 85,000.

5. Find the missing number: $45,000 + \underline{\hspace{2cm}} = 100,000$

SECTION B: REAL WORLD APPLICATIONS

1. Atharv's family traveled 12,450 miles by air and 3,450 miles by car last year. What was their total travel distance?

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2. A library has 85,000 books. If 24,560 books are issued to students, how many books are still in the library?

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3. Priyanka had \$15,000 in her travel fund. She spent \$4,550 on tickets to Florida and \$2,800 on hotels. How much money is left in the fund?

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Chapter 3: Multiplication & Division

SECTION A: CORE CONCEPTS

1. Multiply: 456×24

2. Multiply: $1,205 \times 35$

3. Divide: $8,460 \div 5$

4. Divide 4,524 by 12. Find the quotient and remainder.

5. Simplify: 45×1000

SECTION B: REAL WORLD APPLICATIONS

1. A restaurant prepares 145 Masala Dosas every day. How many dosas will they prepare in a normal year (365 days)?

2. Anaya has 1,248 building blocks. She wants to divide them equally into 8 large storage boxes. How many blocks will go into each box?

3. The cost of one EGO Power+ lawn mower is \$450. What will be the total cost of 15 such mowers?

Chapter 4: Fractions & Decimals

SECTION A: CORE CONCEPTS

1. Write three equivalent fractions for $\frac{2}{5}$.

2. Compare using $<$, $>$ or $=$: $\frac{3}{4}$ ___ $\frac{5}{8}$

3. Add: $\frac{3}{10} + \frac{4}{10}$

4. Subtract: $\frac{7}{8} - \frac{3}{8}$

5. Convert $\frac{45}{100}$ into a decimal.

SECTION B: REAL WORLD APPLICATIONS

1. Atharv ate $\frac{2}{8}$ of a pizza and Anaya ate $\frac{3}{8}$ of the same pizza. How much pizza did they eat in total?

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2. Priyanka bought $\frac{3}{4}$ kg of cherries from the orchard. She used $\frac{1}{4}$ kg to make a pie. How many kilograms of cherries are left?

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3. Mukesh spent 0.45 of his money on books. Write this decimal as a fraction.

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Multi-Concept Combined Practice

Level 1: Core Proficiency (Average Student) - SET 1

1. Write the number 456,789 in words.

2. Round 4,562 to the nearest thousand.

3. Add 45,678 and 32,194.

4. Subtract 18,500 from 50,000.

5. Multiply 345 by 24.

6. Divide 846 by 6.

7. What is the place value of the digit 7 in 8,765,432?

8. Find the perimeter of a square with a side length of 15 cm.

9. Write an equivalent fraction for $\frac{2}{3}$.

10. Add the fractions: $\frac{2}{8} + \frac{3}{8}$.

Multi-Concept Combined Practice

Level 1: Core Proficiency (Average Student) - SET 2

1. Write the expanded form of 105,402.

2. Round 8,921 to the nearest hundred.

3. A warehouse has 25,000 packets of Crunchley makhana. 12,450 are shipped. How many are left?

4. Multiply 506 by 35.

5. Divide 1,245 by 5.

6. Atharv bought a book for \$15.50 and a pen for \$4.25. How much did he spend in total?

7. What fraction of a day is 8 hours?

8. Convert 4500 milliliters to liters.

9. Find the area of a rectangle with length 12m and width 8m.

10. Write $\frac{4}{10}$ as a decimal.

Multi-Concept Combined Practice

Level 2: Advanced Application (Top 10%) - SET 1

1. Priyanka drove 1,245 miles from San Jose. Then she drove another 856 miles. If her goal is 3,000 miles, how far does she still need to drive?

2. Multiply 4,052 by 48.

3. A factory produces 1,450 Jigyasa learning kits every day. How many will they produce in the month of March?

4. Divide 4,524 by 12. What is the quotient?

5. Subtract 45.67 from 100.00.

6. A rectangular garden is 24m long and 15m wide. What is the length of the fence needed to surround it?

7. Simplify the fraction $\frac{16}{24}$ to its lowest terms.

8. Which is larger: $\frac{3}{4}$ or $\frac{5}{8}$? Show your work.

9. Find the sum of the first 5 multiples of 8.

10. Anaya has \$50. She spends \$12.75 on toys and \$8.50 on snacks. How much change does she receive?

Multi-Concept Combined Practice

Level 2: Advanced Application (Top 10%) - SET 2

1. Mukesh ordered 15 boxes of electronics. Each box weighs 45 kg. What is the total weight in grams?

2. A library in Milpitas has 45,600 books. They are distributed equally onto 150 large shelves. How many books are on each shelf?

3. Add $\frac{3}{5}$ and $\frac{1}{10}$. (Hint: Make the denominators the same).

4. A movie at Universal Studios starts at 2:45 PM and lasts for 135 minutes. What time does the movie end?

5. Form the smallest 6-digit odd number using the digits 2, 0, 8, 5, 1, 9.

6. Estimate the product of 487 and 32 by rounding each to the nearest ten first.

7. Subtract $\frac{3}{4}$ from 1 whole.

8. If 1 kilogram of apples costs \$4.50, how much will 3.5 kilograms cost?

9. The area of a rectangle is 120 sq cm. If its length is 15 cm, what is its width?

10. List all the factors of 36.

Multi-Concept Combined Practice

Level 3: Logical Reasoning (Top 1%) - SET 1

1. A tank holds 500 liters of water. It leaks at a rate of 15 liters per hour. How much water will be left in the tank after 24 hours?

2. Multiply 999 by 99 using a mental math shortcut, and show your steps.

3. Atharv reads $\frac{3}{8}$ of a book on Saturday and $\frac{1}{4}$ of it on Sunday. If the book has 240 pages, how many pages are left to read?

4. A digital clock shows 10:15 AM. What is the angle between the hour hand and the minute hand? (Think carefully!)

5. I am a prime number between 40 and 50. If you reverse my digits, I am a multiple of 7. What number am I?

6. Divide 10,000 by 16 and write the remainder as a decimal.

7. A train leaves San Jose at 08:45 and travels at 60 mph. It reaches its destination at 11:15. How far did the train travel?

8. Find the perimeter of a shape made by attaching 3 identical squares in a row, if the area of one square is 25 sq cm.

9. Write the decimal 0.125 as a fraction in its simplest form.

10. How many leap years are there between the year 2001 and 2026?

Multi-Concept Combined Practice

Level 3: Logical Reasoning (Top 1%) - SET 2

1. Priyanka uses an EGO Power+ mower to cut grass. She cuts a lawn of 400 sq meters in 20 minutes. At this rate, how long will it take to cut 1,000 sq meters?

2. Subtract 12.345 from 50.1.

3. Find a fraction that is exactly halfway between $\frac{1}{4}$ and $\frac{1}{2}$.

4. A pattern goes: 2, 6, 12, 20, 30... What is the 10th number in this pattern?

5. A bucket is $\frac{2}{5}$ full of water. When 6 liters are added, it becomes $\frac{7}{10}$ full. What is the total capacity of the bucket?

6. What is the largest 4-digit number that is exactly divisible by 15?

7. If $A \times B = 24$, $B \times C = 48$, and $A \times C = 32$. What are the values of A, B, and C? (Assume they are positive whole numbers).

8. Anaya arranged her building blocks in a solid square grid. She had 15 blocks left over. She tried making the square one block wider and longer, but she was 10 blocks short. How many blocks does she have?

9. Express 45 minutes as a fraction of a day in its lowest terms.

10. The sum of three consecutive odd numbers is 87. What are the numbers?

Multi-Concept Combined Practice

Level 4: Critical Thinking (Top 0.1%) - SET 1

1. A snail is at the bottom of a 20-foot pipe. Every hour, it climbs up 4 feet, but then slides down 2 feet while resting. How many hours will it take to escape the pipe?

2. What is the last digit (units digit) of the number you get when you multiply 3 by itself 20 times?

3. Atharv has \$100. He buys exactly 100 items. Pencils cost \$5, erasers cost \$1, and paperclips cost \$0.05. How many of each did he buy? (He must buy at least one of each).

4. In a tournament of 64 teams, how many total matches must be played to determine a single champion in a single-elimination bracket?

5. How many 3-digit numbers have the sum of their digits equal to 5?

6. A family takes a road trip to Walt Disney World. They travel half the distance at 40 mph and the other half at 60 mph. What is their average speed for the whole trip? (Careful, it's not 50 mph).

7. If you write the numbers from 1 to 100, how many times will you write the digit '9'?

8. The product of the ages of 3 children is 72. The sum of their ages is 14. What are their ages?

9. A cube is painted red on all sides, then cut into 64 smaller identical cubes. How many of the small cubes have exactly two red faces?

10. If 1.5 cats can catch 1.5 mice in 1.5 minutes, how many cats are needed to catch 60 mice in 60 minutes?

Multi-Concept Combined Practice

Level 4: Critical Thinking (Top 0.1%) - SET 2

1. You have a 3-liter jug and a 5-liter jug. How can you measure exactly 4 liters of water? (List the steps).

2. What comes next in the sequence: 1, 11, 21, 1211, 111221, _____?

3. A wooden pole is stuck in the mud. $\frac{1}{3}$ of the pole is in the mud, $\frac{1}{4}$ is in the water, and 5 meters are above the water. How long is the entire pole?

4. If you divide 100 by half, and then subtract 10, what is your answer?

5. At Chaat Bhavan, 3 samosas and 2 lassis cost \$12. 2 samosas and 3 lassis cost \$13. How much does 1 samosa cost?

6. A grandfather clock takes 6 seconds to strike 4 o'clock (4 chimes). How many seconds will it take to strike 8 o'clock?

7. Find the sum of all the angles in a 5-sided polygon (pentagon).

8. I am a 3-digit number. My hundreds digit is twice my tens digit. My ones digit is the sum of my hundreds and tens digit. What numbers could I be?

9. You have 12 coins. One is fake and has a different weight (you don't know if it's heavier or lighter). Using a balance scale exactly 3 times, how do you find the fake coin?

10. Using exactly four 4s and any math symbols, create an equation that equals 20.

Multi-Concept Combined Practice

Level 5: National Olympiad (Top 10) - SET 1

1. A palindrome number reads the same forwards and backwards (e.g., 121). The odometer on Priyanka's car reads 15951. How many more miles must she drive to see the NEXT palindrome number?

2. There are 5 houses of 5 different colors. In each house lives a person of a different nationality. Each person drinks a different beverage, smokes a different brand of cigar, and keeps a different pet. If the Brit lives in the red house, and the Swede keeps dogs... (Provide a logical deduction puzzle strategy for solving this type of grid).

3. Atharv and his brother are painting a fence. Atharv can paint it alone in 3 hours. His brother can paint it alone in 6 hours. If they work together, how long will it take?

4. A train traveling at 60 km/h crosses a 100m bridge in 12 seconds. What is the length of the train?

5. Two numbers have a Greatest Common Divisor (GCD) of 6 and a Least Common Multiple (LCM) of 36. What could the two numbers be?

6. How many squares (of all possible sizes) are there on a standard 8x8 chessboard?

7. If $5^x = 125$, and $2^y = 64$, what is the value of $x * y$?

8. A piece of paper is 0.1 mm thick. If you fold it in half 20 times, roughly how thick will it be? (Just estimate the scale: mm, cm, m, or km?)

9. Find the missing number in the grid: Top row (8, 2, 16), Middle row (9, 3, 27), Bottom row (7, 4, ?)

10. What is the largest number of pieces you can cut a pizza into using exactly 4 straight cuts?

Multi-Concept Combined Practice

Level 5: National Olympiad (Top 10) - SET 2

1. A farmer has cows and chickens. There are 50 heads and 140 legs in the farmyard. How many cows and how many chickens are there?

2. If you multiply all the odd numbers from 1 to 99 together, what is the last digit of the product?

3. A hollow cube has an outer volume of 1000 cubic cm. The walls are 1 cm thick. What is the volume of the empty space inside?

4. In a class, 60% of students like math, 50% like science, and 20% like neither. What percentage of students like BOTH math and science?

5. A magical lily pad doubles in size every day. It covers an entire pond in 30 days. On what day did it cover exactly one-quarter of the pond?

6. What is the angle between the hour and minute hands of a clock at 3:15 exactly?

7. Anaya puts 1 grain of rice on the first square of a chessboard, 2 on the second, 4 on the third, doubling each time. Write an expression for the number of grains on the 10th square.

8. If $A = 1$, $B = 2$... $Z = 26$. What 4-letter word has a product of its letters equal to 455?

9. You have a circular cake. You must cut it into 8 equal pieces using exactly 3 straight cuts. How do you do it?

10. Find a 10-digit number where the first digit indicates how many 0s are in the number, the second indicates how many 1s, the third how many 2s, and so on up to 9.
